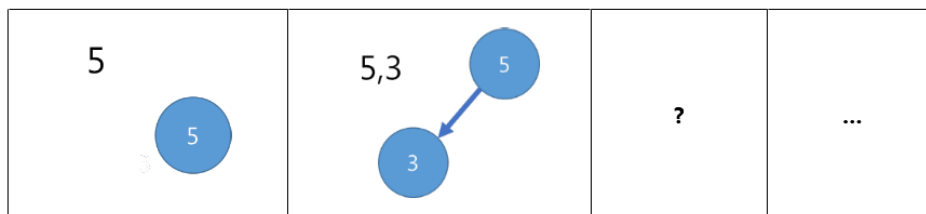


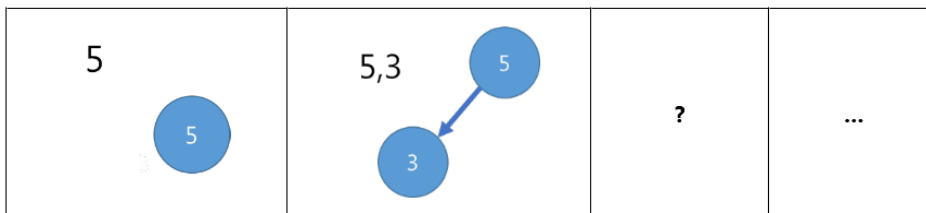
주의사항: 생성형 AI 를 사용하여 과제 수행 (<http://gpt.jbnu.ai> 또는 개인 소유 AI) 2개 (GPT, Gemini, Claude등 임의로 선정)를 사용하여, 각 문제의 **해답** 및 **핵심기능요약**, **문제출제의도** 및 이를 얻기 위한, **입력 프롬프트와 결과, 결과분석, 개인적인 사용 소감** 작성

Home Work #2

1. Illustrate each step of the insert execution of the binary search tree with the following input sequences: (5, 3, 7, 8, 2, 9, 1)



2. Illustrate each step of the delete execution of the binary search tree with the above binary search tree result and the delete sequence: (1, 2, 5, 8, 9).



3. Illustrate each step of the insert execution of the AVL tree with the following input sequence: (5, 3, 7, 8, 2, 9, 1).
4. Illustrate each step of the delete execution of the AVL tree with the above AVL tree result and the delete sequence: (1, 2, 5, 8, 9).
5. Install Python package “bintrees” and write program for the above 4 problems. Show that your 4 illustration results are correct comparing the preorder, inorder and postorder traversal of the illustration results with the preorder, inorder and postorder traversal output of your program. Attach your result.

(Student's ID and Name should be printed in the output.)

Package "bintrees"

Official documentation: <https://pypi.python.org/pypi/bintrees/2.0.2> (Read this to use methods)

(You can use the recent version of bintrees)

Code guide

```
from bintrees import BinaryTree
from bintrees import AVLTree

# construct tree
btree = BinaryTree()
avltree = AVLTree()
# how to insert 3
# tree.insert(key,value)
btree.insert(3, None)
# use the key argument, the value argument is not important in our HW
#read https://pypi.python.org/pypi/bintrees/2.0.2 to use various method for HW
write your program
```

Output format (Fill "?"s and attach the result)

Python index.py

1.

In-order ? ? ? ? ? ? ?

Pre-order ? ? ? ? ? ? ?

Post-order ? ? ? ? ? ? ?

2.

In-order ? ?

Pre-order ? ?

Post-order ? ?

3.

In-order ? ? ? ? ? ? ?

Pre-order ? ? ? ? ? ? ?

Post-order ? ? ? ? ? ? ?

4.

In-order ? ?

Pre-order ? ?

Post-order ? ?